

DevOps Engineer Roadmap

1. Foundations

- Learn Linux basics and shell scripting
- Understand networking fundamentals (TCP/IP, DNS, HTTP/HTTPS)
- Understand OS Concepts, Permissions, Process Management

2. Programming & Scripting

- Bash scripting
- Python basics

3. Version Control

- Git basics (commit, push, pull, branching)
- Git advanced concepts (rebasing, merge strategies)

4. CI/CD

- Jenkins, GitHub Actions, GitLab CI
- Build pipelines, automated testing

5. Containers

- Docker basics & architecture
- Docker images, containers, Docker Compose

6. Container Orchestration

- Kubernetes fundamentals
- Deployments, Services, Ingress, ConfigMaps, Secrets

7. Cloud Platforms

- AWS (EC2, S3, IAM, VPC, Lambda)
- Azure or GCP basics

8. Infrastructure as Code

- Terraform (main tool to learn)
- CloudFormation (optional)

9. Monitoring & Logging

- Prometheus & Grafana
- ELK / EFK Stack

10. Security in DevOps

- Secrets management (Vault)
- Basic cloud and network security

11. Additional Tools

- Ansible (Configuration management)
- Helm (Kubernetes package manager)

12. Soft Skills

- Communication, debugging, documentation, collaboration